

SOURCE

Maintained supply
Dispatch D
Shortage signal

PENDING Q

Already dispatched
Arrival $A = Q / \tau$

RECEIVER X

Usable stock
Consumption h

Declared feedback: $D = \nu (L - X)$

ELIMINATE THE PENDING DEVIATION q

Visible response = initial-history term + causal memory integral

$K(r) = -(\nu / \tau) \exp(-r / \tau)$, for $r \geq 0$

$q(0) = 0$ means the operating pipeline stock, not an empty pipe.

Explicit pending state and its memory representation are equivalent.

Deleting both would lose commitments already made.